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PROFESSIONAL SUMMARY

Senior Software & Machine Learning Engineer with a proven track record of designing, scaling, and operating business-critical distributed systems and production-ready applied ML solutions. Specialise in the engineering intersection of high-throughput, real-time data-ingestion pipelines and live ML inference systems, processing high-dimensional, noisy data under strict operational constraints. Expert in architecting resilient, event-driven microservices (CQRS, DDD) across multi-cloud environments (GCP, Azure) and managing the end-to-end ML lifecycle from rigorous experiment design to deployment. Thrives in high-autonomy, fast-moving environments, combining strong architectural ownership with direct stakeholder collaboration to deliver reliable, production-grade intelligence at scale.

TECHNICAL SKILLS

Core Programming: Python, C#/.NET, TypeScript, JavaScript

Machine Learning & Digital Signal Processing: TensorFlow, Keras, Scikit-learn, NumPy, SciPy, Pandas, Sequence Modelling, Deep Learning, Wavelet Decomposition, Spectral Analysis, Time-Frequency Transformations

Backend & Distributed Systems: Event-Driven Microservices, Domain-Driven Design (DDD), CQRS, Distributed Systems Design, RESTful APIs, Asynchronous Workflows, OAuth2 / OIDC (IdentityServer4)

Cloud & DevOps: GCP (Pub/Sub, Spanner, BigQuery, Cloud Build), Azure (Event Hubs, Service Bus, DevOps, Blob Storage), Docker, Kubernetes, CI/CD Pipelines

Databases & Storage: PostgreSQL, SQL Server, Elasticsearch, Redis (Caching & Messaging), Database Sharding & Partitioning Architecture

Engineering Practices: Agile/Scrum SDLC, Test-Driven Development (TDD), Unit & Integration Testing, System Observability (Kibana), Cross-Functional Stakeholder Communication

EXPERIENCE

Applied ML Engineer (Biomedical AI) | ABSPIE Lab (University of Warwick), 2021 - 2025

Developed advanced physiological monitoring and forecasting methods for deployment in wearable medical-device systems, working with high-dimensional, noisy, multimodal biomedical datasets. Designed deep-learning models, built robust end-to-end signal-processing pipelines, and evaluated model performance using clinically relevant metrics within a multidisciplinary medical engineering research lab.

- **Deep Learning & Sequence Modelling:** Designed and evaluated deep-learning architectures for physiological monitoring and advanced signal processing; built and validated models for classification, regression, and sequence-modelling tasks, utilising unsupervised representation learning for feature extraction across multiple biosignal modalities
- **Advanced Signal Processing Pipelines:** Engineered robust, end-to-end preprocessing pipelines for large-scale physiological datasets; developed a novel wavelet-based decomposition filtering methodology alongside custom segmentation and data-cleaning algorithms to support downstream modelling
- **Statistical Evaluation & Robustness:** Conducted rigorous data exploration, statistical analysis, and experimental evaluation on original physiological datasets; utilized clinically relevant metrics to assess model robustness, generalisability, and real-world relevance
- **Cross-Functional Translation:** Collaborated with clinicians, researchers, and industry partners to translate technical findings into practical physiological-monitoring applications aligned with commercial and clinical requirements
- **International Recognition & Awards:** Awarded Runner-Up for the **IFMBE Young Investigator Award (2023)** for outstanding medical engineering research; synthesized and presented applied deep-learning frameworks for continuous ambulatory physiological monitoring at international conferences and published findings in peer-reviewed journals

Tech: Python, TensorFlow, Scikit-learn, NumPy, SciPy, Pandas

Software Engineer | MandM Direct 2020 - 2021

Contributed to the end-to-end migration of legacy on-premise systems to a fully cloud-native, event-driven architecture, resolving major scaling and operational constraints across the business. Implemented core components of a retail order-processing platform, real-time data-ingestion pipelines, and high-availability services for peak-traffic periods, while also developing internal tooling to improve visibility and decision-making across the organisation.

- **High-Availability Order Processing:** Designed and developed a cloud-based retail order-processing platform built on event-driven microservices, enabling scalable and resilient handling of high-volume retail transactions; engineered fault-tolerant components and implemented Redis-backed caching and message-based workflows to maintain performance and continuity during peak-traffic events
- **GCP Data Ingestion Pipeline:** Built a GCP-native data ingestion engine using Pub/Sub, Spanner, and BigQuery to deliver order, inventory, and customer events in near real-time for downstream processing by business intelligence and analytics
- **Legacy Integration & Workflows:** Integrated new cloud microservices with legacy enterprise systems via real-time data synchronization pipelines, maintaining system consistency with zero business downtime to support continuous operations
- **Internal Tooling:** Developed operational dashboards using Angular and TypeScript to support cross-departmental operational visibility and data-driven decision-making
- **Team Contribution & Mentorship:** Mentored junior developers through code reviews, pair programming, and architectural guidance, while supporting team leadership by contributing to sprint planning and backlog refinement

Tech: C#/.NET, TypeScript, Angular, GCP (Spanner, BigQuery, Pub/Sub), Redis

Founding Software Engineer | Bioepic 2017–2019

Early employee and first dedicated backend engineer at a wellness-focused health-tech startup, owning backend architecture, ML inference, and high-volume data ingestion from the initial prototype phase through production as the company scaled from a small founding team to 80+ staff. Collaborated with data science, product, and clinical teams to support validation studies and the company's transition toward medical-device regulatory certification.

- **Architectural Scaling & CQRS:** Decoupled ingress from egress at the service layer to support an anticipated 10M daily active users; isolated PostgreSQL as a high-speed engine for write-heavy raw data, while routing ML inference outputs and metadata to Elasticsearch, eliminating data contention bottlenecks
- **ML Infrastructure & Ingestion:** Designed and managed a cloud platform powering real-time ML inference on wearable data; established CI/CD pipelines to deploy versioned models to ensure consistent, reliable production behaviour, built secure APIs and high-throughput ingestion pipelines for multimodal physiological data, using Azure Event Hubs, ensuring reliable delivery and processing under variable network conditions and device configurations
- **Data Architecture & Partitioning:** Key contributor to the core migration from SQL Server to PostgreSQL, executed specifically to scale ingestion throughput and natively support unstructured data structures via optimized JSON-backed querying and indexing; contributed to database sharding and partitioning strategies to support growing data volumes and improve query performance
- **Event-Driven Microservices:** Implemented containerised, asynchronous microservices behind an API Gateway powered by Azure Service Bus; applied Domain-Driven Design (DDD) principles and bounded contexts to establish clear service boundaries, and utilised test-driven development (TDD) to ensure safe iteration.
- **Platform Security (OAuth2/OIDC):** Engineered modern authentication and authorization across .NET Core microservices using IdentityServer4, implementing JWT-based API protection, scoped access control, and claims-driven service-to-service security
- **Cloud, Containers & CI/CD:** Containerized services with Docker, deployed them to Kubernetes clusters, and established CI/CD pipelines via Azure DevOps to automate deployments and maintain operational consistency
- **Observability & Search:** Developed Elasticsearch-backed search and indexing capabilities to support rapid querying of time-series metadata, and built Kibana dashboards to provide real-time operational and business insights across departments

Tech: C#/.NET Core, ASP.NET Core Web API, PostgreSQL, Elasticsearch, Docker, Kubernetes, Azure Service Bus, Azure Event Hubs, Azure DevOps, IdentityServer4, Kibana

Full-Stack Developer | Franklin Hodge Industries 2015 – 2017

Sole developer responsible for designing, building, and modernising internal systems across engineering, manufacturing, and commercial teams. Partnered directly with stakeholders to translate operational requirements into clear technical requirements to deliver high-impact solutions end-to-end.

- **Data Infrastructure:** Led the migration of core company data from a single shared Microsoft Access file to a relational SQL Server database, including schema design, constraints, indexing, and full migration tooling resulting in improved performance, reliability, and data integrity
- **Workflow Automation:** Built custom document-generation and automation pipelines integrated with industry-standard software, reducing manual workload, improving data accuracy, and increasing operational efficiency across engineering and commercial teams
- **Web Applications:** Designed, developed, and deployed a web-based CRM platform to streamline purchase-order processing, sales workflows, and customer management operations
- **Legacy Refactoring:** Modernised business-critical legacy internal applications through structured refactoring for clearer data-flow design and improved maintainability

Tech: C#/.NET Framework, ASP.NET MVC, SQL Server, JavaScript, jQuery, IIS

EDUCATION & AWARDS

BSc (Hons) Theoretical Physics | Swansea University 2014

IFMBE Young Investigator Award (Runner-Up) | International Federation for Medical and Biological Engineering 2023